The DNA of Numbers

$p \equiv 1 \pmod{q}$

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A theory of digital roots, Diophantine representations,
fractal ladders and the inevitable three-way division of moduli.

Foreword

Every number can be classified into three classes of moduli:

1. The class in which the digital root traverses all 9 levels.
2. The class in which the digital root remains trapped in {1, 4, 7}.
3. The class in which the digital root is constantly 1.

Dataset for the moduli 2 through 900.

Class A is the largest group. This is because most numbers are not divisible by 3. Class B is the intermediate class: the multiples of 3 that are not multiples of 9. Class C is the smallest group: the multiples of 9.

The ratio of thirty numbers across the three classes is (20 : 7 : 3). The ratio of 90 numbers is 60 : 20 : 10 = 6 : 2 : 1. This ratio 6 : 2 : 1 is no coincidence. It is a direct consequence of divisibility by 3 and 9.

Class A (6/9): the numbers that are not divisible by 3, i.e. 2/3 of all numbers.

Class B (2/9): the numbers that are divisible by 3 but not by 9.

Class C (1/9): the numbers that are divisible by 9.

The ratio 6 : 2 : 1 is exactly the same as the ratio for moduli 2 through 900 (599 : 200 : 100 = 6 : 2 : 1). The numbers 3 and 9 were not chosen. They follow inevitably from the structure of the digital root.

The digital root works with 9

The digital root is defined as $dr(n) = 1 + ((n - 1) \bmod 9)$. The modulus is 9. That is not a choice. It is a direct consequence of the fact that we work in the decimal system (10 digits, and $10 \equiv 1 \bmod 9$). Thus 9 is the natural modulus of our number system.

The +q-translation on $\mathbb{Z}/9\mathbb{Z}$

In this system $N = p \cdot a + q \cdot b$ we examine what happens when we add q to the digital root. The operation +q on the set $\mathbb{Z}/9\mathbb{Z}$ is a translation. The question is: how many steps does it take before we return to the starting point? That is the orbit length of the translation.

The orbit length depends on the greatest common divisor of q and 9.

If gcd(q, 9) = 1 (q not divisible by 3), then the translation traverses all 9 values. That is Class A (The Circle).

If gcd(q, 9) = 3 (q divisible by 3, not by 9), then the translation traverses 3 values (1, 4, 7). That is Class B (The Arc).

If gcd(q, 9) = 9 (q divisible by 9), then the translation remains on 1 value (d = 1). That is Class C (The Ray).

The numbers 3 and 9 are the divisors of 9. Because the digital root is based on 9, the divisors of 9 (1, 3 and 9) are the only possible orbit lengths. The three-way division is therefore not imposed. It is a direct consequence of the question: how many times must you add q before you return to the same digital root?

The ratio 6 : 2 : 1 is likewise inevitable, because it follows exactly from divisibility by 3 and 9. These are mathematical regularities, not design choices. The structure already resides in the numbers. I have only made it visible.

Glossary: symbols, concepts and notation

This glossary contains all symbols and concepts used in the manuscript, arranged by category. It forms the methodological foundation of the theory.

Numbers and sets

$\mathbb{Z}$ (Integers)
the set of all integers (…, −2, −1, 0, 1, 2, …). Example: $5 \in \mathbb{Z}$.

$\mathbb{Z}_{\geq 0}$ (Non-negative integers)
the set of 0, 1, 2, 3, … Example: $0 \in \mathbb{Z}_{\geq 0}$.

$\mathbb{N}$ (Natural numbers)
the set of 1, 2, 3, … Example: $3 \in \mathbb{N}$.

$\in$ / $\notin$ (Element of / Not an element of)
“belongs to the set” / “does not belong to the set”. Example: $5 \in \mathbb{Z}$, $-1 \notin \mathbb{Z}_{\geq 0}$.

{…} (Set)
a group of numbers between braces. Example: {1, 4, 7}.

The capital letters of the equation

N (Number)
the number under investigation. Example: N = 958.

p (Position)
the first step, a positive integer. Example: p = 19.

q (Quantum / modulus)
the second step, a positive integer. Example: q = 9.

a (Amount)
the first coordinate, how many times p fits into N. Example: a = 4.

b (Balance)
the second coordinate, how many times q fits into N. Example: b = 98.

a_0 (Starting amount)
the smallest non-negative value of a, equal to the digital root of N. Example: $a_0 = 4$. For multiples of 9 we have $a_0 = 0$, while dr = 9.

a_{max} (Maximum amount)
the largest value of a on the ladder. Example: $a_{max} = 49$.

b_{min} (Minimum balance)
the b-value corresponding to a_{max}. Example: $b_{min} = 3$.

b_{max} (Maximum balance)
the b-value corresponding to a_0. Example: $b_{max} = 98$.

R (Representations)
the number of valid pairs (a, b). Example: R(958) = 6.

k (Ladder step)
the step index on the fractal ladder. Example: k = 2.

Γ(k) (Family threshold)
the smallest number with at least k representations. Example: Γ(10) = 1683.

Γ_leap(k) (Leap threshold)
playful analogy for the leap-year cycle. Example: 36k − 4.

m (Level)
the number of steps or rounds in a family or on the spiral. Example: m = 2, with p = 1 + q·m.

t (Time step)
the number of steps in a cycle. Example: t = 9.

The digital root and the classes

d (Spiral height)
the height of the spiral or the digital root of p. Example: d = 4. Multiples of nine yield 9, not 0.

dr(N) (Digital root)
the function that reduces the digit sum to a single digit: dr(n) = 1 + ((n − 1) mod 9). Example: dr(958) = 4.

dr_q(N) (Generalised digital root)
the digital root based on modulus q: dr_q(N) = 1 + ((N − 1) mod q). Values between 1 and q.

S(N) (Digit sum)
the sum of all digits of N, once. Example: S(958) = 22.

A (The Circle / Class A)
the class of moduli in which the digital root traverses all 9 levels ($q \not\equiv 0 \bmod 3$). Example: q = 7.

B (The Arc / Class B)
the class of moduli in which the digital root remains trapped in {1, 4, 7} ($q \equiv 0 \bmod 3$, $q \not\equiv 0 \bmod 9$). Example: q = 60.

C (The Ray / Class C)
the class of moduli in which the digital root is constantly 1 ($q \equiv 0 \bmod 9$). Example: q = 3600.

Residue class
the d-value class $[N] \in \mathbb{Z}/9\mathbb{Z}$, the algebraic object. Example: [958] = [4].

α (Cycle anchor)
the pair (n, n+1) where n ≡ 0 (mod 9) and n+1 ≡ 1 (mod 9). Example: 9000 → 9001.

κ (Clock constant)
the LCM of 6 and 7, i.e. 42, which forms the bridge between the time units and the Diophantine equation.

Operations and relations

mod (Modulus)
the remainder upon division. Example: 958 mod 9 = 4.

≡ / ≢ (Congruent / Not congruent)
has the same remainder upon division / does not have the same remainder. Example: 19 ≡ 1 (mod 9).

gcd(a, b) (Greatest common divisor)
the largest number that divides both a and b. Example: gcd(4, 9) = 1.

LCM(a, b) (Least common multiple)
the smallest number divisible by both a and b. Example: LCM(4, 9) = 36.

⌊x⌋ / ⌈x⌉ (Floor / Ceiling function)
rounding down / rounding up. Example: ⌊132.63⌋ = 132, ⌈132.63⌉ = 133.

Σ (Summation)
the sum of a series of numbers. Example: Σ i = 1+2+3 = 6, from i=1 to 3.

Δ (Delta)
the difference between two consecutive values. Example: Δdr = +1.

⇔ / ⇒ / ⟷ (Equivalence / Implication / Correspondence)
“if … then …”, or “two things are the same, viewed differently”. Example: d ⟷ a_0.

The fractal ladder

L (Fractal ladder)
the ordered sequence of representations (a, b) of N, where a increases by q and b decreases by p. Example: L(958).

L_k (Rung)
the k-th representation on the fractal ladder. Example: L_2 is (22, 60).

Digit reduction
the iterative digit sum that leads to a_0 or dr(N). Example: 958 → 22 → 4.

Matryoshka principle
every value in the reduction chain N → S(N) → … → dr(N) appears as an a-coordinate in the representations of N.

Chapter 1: the core formula and the digital root

1.1 Digital root

Every number has a starting point (a_0). The digital root tells us exactly what the smallest coordinate (a_0) is. Consequently the end point (a_{max}) in the representations can be derived exactly, because the coordinate a increases by q at every step or rung and the coordinate b decreases by p. We can therefore also compute the number of rungs (R) with an exact, closed formula. With this knowledge we can derive the highest value (a_{max}). Because the total sum (N) is fixed, the minimal balance (b_{min}) follows at once.

Example: N = 958, p = 19, q = 9, a_0 = 4, a_{max} = 49, b_{min} = 3, R = 6.

Note the convention for multiples of 9: N mod 9 = 0, while dr(N) = 9. In the formulae we use a_0 = N mod q, so a_0 = 0. The digital root remains 9. Both values, 0 and 9, appear on the ladder when relevant.

The smallest coordinate a_0 is the digital root of N. The digital root can be computed in two ways.

Via remainder division: divide N by 9 and take the remainder. Example: 958 ÷ 9 = 106 remainder 4, so a_0 = 4.

Via digit reduction: add the digits until a single digit remains. Example: 9+5+8 = 22, then 2+2 = 4, so a_0 = 4.

Both methods yield the same number. The formal definition dr(n) = 1 + ((n − 1) mod 9) is the neat mathematical transcription of digit reduction. The "1 + (…)" ensures that the result always lies between 1 and 9. For multiples of 9 this formula yields 9 instead of 0. For the coordinate a_0 we use the remainder representative, so for multiples of 9 we have a_0 = 0. Furthermore a_0 is always equal to dr(N).

Example: dr(958) = 1 + ((958 − 1) mod 9) = 1 + (957 mod 9) = 1 + 3 = 4. And indeed 958 mod 9 = 4, so a_0 = 4.

1.2 The linear Diophantine equation

A Diophantine equation is an equation in which we accept only integers as solutions. Our main equation is N = p·a + q·b. The principal condition is p ≡ 1 (mod q). This means that p divided by q must leave remainder 1. For p = 19 and q = 9 we have: 19 ≡ 1 (mod 9).

Example: 958 = 19·4 + 9·98. Here a = 4 and b = 98.

1.3 The bounds of a: a_0 and a_{max}, and the exact R-formula

Because p ≡ 1 (mod q), we have N = p·a + q·b ≡ a (mod q). The remainder of N upon division by q is therefore equal to the remainder of a. We now determine both the starting point (a_0) and the end point (a_{max}) of the ladder.

The starting point (a_0) is determined via two routes that yield the same result.

Route 1, remainder division: divide N by q and take the remainder. Example N = 958: 958 mod 9 = 4, so a_0 = 4.

Route 2, digit reduction: add the digits until you reach a valid a-value. Example: 9+5+8 = 22, then 2+2 = 4. Both values, 22 and 4, are obligatory coordinates on the ladder L (Matryoshka principle).

Both routes yield the same a_0, because a_0 is the digital root of N. Remainder division is the direct computational method. Digit reduction additionally supplies an obligatory intermediate station on the ladder.

The exact R-formula for the number of representations is:

$$R(p, q; N) = \lfloor(\lfloor N/p \rfloor - a_0) / q\rfloor + 1, \text{ provided } N \geq p{\cdot}q.$$

Note: if N < p·q, i.e. N < 171 for p = 19 and q = 9, then the representation (0, N/q) must be added explicitly when N ≡ 0 (mod q). This is the boundary case.

Example N = 958: ⌊958/19⌋ = 50 and a_0 = 4, so R = ⌊(50 − 4)/9⌋ + 1 = ⌊46/9⌋ + 1 = 5 + 1 = 6. Then a_{max} = a_0 + q·(R − 1) = 4 + 9·5 = 49. And b_{min} = (N − p·a_{max})/q = (958 − 19·49)/9 = 3.

1.4 The bridge between spiral and representation

The digital root is the same function in both worlds: d = 1 + ((p − 1) mod 9) ⟷ a_0, with the 0/9 convention. In the spiral theory d determines the height. In the representation theory a_0 determines the smallest coordinate. It is the same quantity, viewed twice.

1.5 The role of the parameter m

p = 1 + q·m. The parameter m is the quotient, the level, not the remainder. For p = 19 and q = 9 we have m = 2, because 2·9 + 1 = 19. The remainder modulo 9 is always 1.

Chapter 2: the three classes (A, B, C)

2.1 What is a class?

A class is a group of moduli that behave in the same way. Depending on the modulus q the behaviour of the digital root falls into three classes.

2.2 The three classes

Class A (The Circle): q ≢ 0 (mod 3). The digital root traverses a full cycle of 9. Examples: q = 2, 4, 5, 7, 8, 10, 365.

Class B (The Arc): q ≡ 0 (mod 3), but q ≢ 0 (mod 9). The digital root makes a short cycle of 3, trapped in {1, 4, 7}. Examples: q = 3, 6, 12, 15, 30, 60, 120, 240.

Class C (The Ray): q ≡ 0 (mod 9). The digital root is constant (d = 1) and the spiral is flat. Examples: q = 9, 18, 27, 36, 360, 3600, 86400.

Scientific formulation: the operation +q on $\mathbb{Z}/9\mathbb{Z}$ is a translation. The orbit length is 9 (Class A), 3 (Class B) or 1 (Class C), depending on gcd(q, 9).

2.3 What do the classes mean in practice?

Class A: the digital root traverses all nine levels, for example the week with q = 7.

Class B: the digital root remains trapped in {1, 4, 7}, for example 60 minutes.

Class C: the digital root remains constantly 1, for example 360° or 3600 seconds.

2.4 Relation to the residue class

We make a strict distinction between the residue class ($[N] \in \mathbb{Z}/9\mathbb{Z}$), the digital root (the representative in {1, …, 9}) and Class A/B/C (the orbit class under the +q-translation).

2.5 The general formula for every modulus q

The theory is general. It works for every modulus q, as long as $p \equiv 1 \pmod{q}$. The three classes are determined by the divisibility of q by 3 and 9, i.e. by gcd(q, 9). The generalised digital root is:

$$dr_q(N) = 1 + ((N - 1) \bmod q)$$

This function yields a value between 1 and q. The A/B/C classification, however, always concerns the behaviour of the ordinary digital root (mod 9) under addition of q.

Example q = 12 (Class B, p = 13): gcd(12, 9) = 3, so orbit length 3 on $\mathbb{Z}/9\mathbb{Z}$. The values of dr_12 are periodic with period 12, but the relevant cycle for the theory is the orbit of the ordinary digital root under +12.

Chapter 3: the family 1675 to 1683 and the threshold

3.1 The nine consecutive numbers

The nine consecutive numbers 1675 through 1683 form the first contiguous block in which each of the nine digital roots 1 through 9 occurs exactly once, and in which every number has reached its full representation count, except one: 1682.

For each number we determine dr, a_0, R, a_{max} and b_{min}. We then verify with $19 \cdot a_{max} + 9 \cdot b_{min}$ whether we recover N.

1675: dr = 1, a_0 = 1, R = 10, a_{max} = 82, b_{min} = 13. Check: 19·82 + 9·13 = 1558 + 117 = 1675.
1676: dr = 2, a_0 = 2, R = 10, a_{max} = 83, b_{min} = 11. Check: 19·83 + 9·11 = 1577 + 99 = 1676.
1677: dr = 3, a_0 = 3, R = 10, a_{max} = 84, b_{min} = 9. Check: 19·84 + 9·9 = 1596 + 81 = 1677.
1678: dr = 4, a_0 = 4, R = 10, a_{max} = 85, b_{min} = 7. Check: 19·85 + 9·7 = 1615 + 63 = 1678.
1679: dr = 5, a_0 = 5, R = 10, a_{max} = 86, b_{min} = 5. Check: 19·86 + 9·5 = 1634 + 45 = 1679.
1680: dr = 6, a_0 = 6, R = 10, a_{max} = 87, b_{min} = 3. Check: 19·87 + 9·3 = 1653 + 27 = 1680.
1681: dr = 7, a_0 = 7, R = 10, a_{max} = 88, b_{min} = 1. Check: 19·88 + 9·1 = 1672 + 9 = 1681.
1682: dr = 8, a_0 = 8, R = 9, a_{max} = 80, b_{min} = 18. Check: 19·80 + 9·18 = 1520 + 162 = 1682. (anomaly)
1683: dr = 9, a_0 = 0, R = 10, a_{max} = 81, b_{min} = 16. Check: 19·81 + 9·16 = 1539 + 144 = 1683.

From 1675 to 1681, b_{min} decreases neatly by 2 at each step. The gap at 1682, with R = 9 instead of 10, translates into a lower a_{max} and a higher b_{min}.

3.2 Why 1682 has one representation fewer

The family threshold for k = 10 is $\Gamma(10) = p \cdot q \cdot (k-1) + (p-1) \cdot (q-1) = 171 \cdot 9 + 18 \cdot 8 = 1539 + 144 = 1683$. From 1683 onwards every number has at least 10 representations. The number 1682 lies one below this threshold and therefore has R = 9.

3.3 Vertical symmetry: a = b and the power of 28

At 1680 there is a full symmetry point: (60, 60). If a = b, then $N = 19 \cdot a + 9 \cdot a = 28 \cdot a$. Every multiple of 28 has a perfect symmetry point on the fractal ladder. 28 = p + q connects the (19, 9)-system with the dualistic week (2 × 14). 2520, the LCM of 1 through 10, has the symmetry point (90, 90).

Chapter 4: the fractal ladder (L)

4.1 What is the fractal ladder?

When we write a number N as $N = p \cdot a + q \cdot b$, there are multiple solutions. The fractal ladder (L) is the insight that these representations are rungs of a ladder. Every rung has the same digital signature, but the coordinates change stepwise: a increases by q per step, and b decreases by p per step.

4.2 Full derivation of N = 958

Step 1: a_0 via digit reduction: 958 → 22 → 4. So $a_0 = 4$.

Step 2: $R = \lfloor(\lfloor 958/19 \rfloor - 4)/9 \rfloor + 1 = \lfloor(50 - 4)/9 \rfloor + 1 = \lfloor 46/9 \rfloor + 1 = 5 + 1 = 6$. Then $a_{max} = 4 + 9 \cdot 5 = 49$.

Step 3: $b_{min} = (958 - 19 \cdot 49)/9 = 3$ and $b_{max} = (958 - 19 \cdot 4)/9 = 98$.

The complete ladder L(958):

L_0 = (4, 98), dr(a) = 4, dr(b) = 8.
L_1 = (13, 79), dr(a) = 4, dr(b) = 7.
L_2 = (22, 60), dr(a) = 4, dr(b) = 6. This is the scan rung, with 22 from the digit reduction.
L_3 = (31, 41), dr(a) = 4, dr(b) = 5.
L_4 = (40, 22), dr(a) = 4, dr(b) = 4. This is a digital symmetry point.
L_5 = (49, 3), dr(a) = 4, dr(b) = 3.

4.3 The general formula for every number N

Step 1: determine $a_0 = N \bmod q$, or via digit reduction.

Step 2: determine $R = \lfloor(\lfloor N/p \rfloor - a_0)/q \rfloor + 1$, provided $N \geq p \cdot q$.

Step 3: determine $a_{max} = a_0 + q \cdot (R - 1)$.

Step 4: determine $b_{min} = (N - p \cdot a_{max})/q$ and $b_{max} = (N - p \cdot a_0)/q$.

Step 5: the ladder runs from $L_0 = (a_0, b_{max})$ to $L_{R-1} = (a_{max}, b_{min})$.

Matryoshka principle: every value in the reduction chain $N \to S(N) \to \ldots \to dr(N)$ appears as an a-coordinate in the representations of N. Proof: because $p \equiv 1 \pmod q$ every digit sum has the same remainder modulo q as the original number. Therefore every intermediate station falls automatically into the a-sequence.

4.4 The vertical axis: the a = b line

Besides the horizontal staircase from a_0 to a_{max} there is a vertical axis: the line where a = b. Because $N = 28 \cdot a$ this line intersects the ladder exactly at the multiples of 28. For N = 2520 the intersection lies at a = 90 and b = 90. This proves that the representations form a perfect geometric grid.

Chapter 5: the Matryoshka structure

When two numbers add up, their representations also add up. This is the Matryoshka structure: one layer, the sum, contains the other layers, the parts.

Matryoshka theorem: if a + b = c, then ($A_1 + A_2$, $B_1 + B_2$) is a valid representation of c. Proof: $p(A_1 + A_2) + q(B_1 + B_2) = (p \cdot A_1 + q \cdot B_1) + (p \cdot A_2 + q \cdot B_2) = a + b = c$. □

This holds on three levels simultaneously: the numbers themselves, their coordinates and their digital roots. Examples: 366 + 592 = 958, 589 + 369 = 958 (an anagram of 958), and the general three-level additivity for arbitrary triples a + b = c.

Chapter 6: perfect numbers and the digital root 1

A perfect number is equal to the sum of its proper divisors, for example 6 = 1+2+3 and 28 = 1+2+4+7+14. Theorem: every even perfect number $P \geq 28$ satisfies $P \equiv 1 \pmod 9$. Consequence: dr(P) = 1 and $a_0 = 1$. Example: $28 = 19 \cdot 1 + 9 \cdot 1$.

Chapter 7: the clock, time units and their classes

The theory of digital roots and the three classes is the building block of our time measurement. The clock is a fractal construction of numbers.

7.1 Class C (The Ray, d = 1)

Hour in seconds (q = 3600), day in seconds (q = 86400), 360-degree circle (q = 360), millennium calendar (q = 360 000), 7000-year cycle (q = 2 520 000) and quarter (q = 90). All these moduli are divisible by 9 and therefore fall into Class C.

7.2 Class B (The Arc, trapped in {1, 4, 7})

Second, minute and hour (q = 60), quarter-hour (q = 15), half-hour and month (q = 30), two hours (q = 120), four hours (q = 240), 12-hour clock (q = 12), day (q = 24) and leap cycle (q = 1461). All these moduli are divisible by 3 but not by 9 and therefore fall into Class B.

7.3 Class A (The Circle)

Millisecond (q = 1000), week (q = 7), two weeks (q = 14), four weeks (q = 28), year (q = 365) and two years (q = 730). All these moduli are not divisible by 3 and therefore fall into Class A.

7.4 The dataset, moduli 2 through 900

For all moduli from 2 through 900 the spiral structure has been analysed, with m = 1 through 10. Result: Class A 599, Class B 200, Class C 100. The count is independent of m.

Chapter 8: the leap-year cycle and the 36-year cycle

The Gregorian leap rule: divisible by 4, except century years that are not divisible by 400. Theorem: gcd(4, 9) = 1, so LCM(4, 9) = 36 years. The full cycle, with the 400-year rule, yields LCM(400, 9) = 3600 years.

Chapter 9: the calibration points 9000 and 9001

9000 (Class C, $a_0 = 0$) to 9001 (Class A, $a_0 = 1$) is the transition from a closed to an open cycle. This is a fractal anchor: the same structure repeats at every scale. For N = 100 we have $9 \cdot 9 + 1 \cdot 19 = 100$. For N = 1000 we have $90 \cdot 9 + 10 \cdot 19 = 1000$. For N = 10 000 we have $900 \cdot 9 + 100 \cdot 19 = 10\,000$.

Chapter 10: the conceptual bridge

This chapter draws connections between the (19, 9)-system and the time units. They are observations. They show that the numbers cohere, but do not constitute proofs that this is more than a consequence of the building blocks 2, 3, 6, 7 and 9.

The LCM sequence around κ = 42 = LCM(6, 7) scales with powers of 10 without changing class. 42, 420, 4200 and 42000 all remain in Class B. Via the factor 4 one obtains $168 = 6 \cdot 19 + 6 \cdot 9 = 28 \cdot 6 = 7 \cdot 24$, the number of hours in a week. Further, $1140 = 19 \cdot 60$ and $2520 = 42 \cdot 60$ connect the p-value and κ in the minute. Finally: $19^2 = 361 = 360 + 1$, the first rung of the flat spiral with q = 360, Class C.

Chapter 11: formulae and results

1. $dr(n) = 1 + ((n - 1) \bmod 9)$, the canonical formula.
2. $a_0 = N \bmod q$, with the convention $a_0 = 0$ for multiples of 9.
3. $R(p, q; N) = \lfloor (\lfloor N/p \rfloor - a_0)/q \rfloor + 1$, the number of representations, provided $N \geq p \cdot q$.
4. $a_{max} = a_0 + q \cdot (R - 1)$.
5. $b_{min} = (N - p \cdot a_{max})/q$ and $b_{max} = (N - p \cdot a_0)/q$.
6. $\Gamma(k) = p \cdot q \cdot (k - 1) + (p - 1) \cdot (q - 1)$, the family threshold.
7. Matryoshka: every digit sum in the chain is a valid a-coordinate.
8. Matryoshka additivity: if a + b = c, then (A_1+A_2, B_1+B_2) counts.
9. Even perfect numbers $P \geq 28$ have dr(P) = 1 and $a_0 = 1$.

10. Leap-9 cycle: LCM(4, 9) = 36 years, fully LCM(400, 9) = 3600 years.
11. Circle root: 19·19 = 361 = 1 + 360, with m = 1 on q = 360, Class C.
12. Generalised digital root: dr_q(N) = 1 + ((N − 1) mod q).

Chapter 12: conclusions

1. The +q-operation is a translation on $\mathbb{Z}/9\mathbb{Z}$. The orbit length is determined by gcd(q, 9).
2. The three classes A/B/C follow formally from the divisibility of q by 3 and 9.
3. Residue class and digital root do not always coincide, as with 0 versus 9.
4. a_0 can be found via two routes. Digit reduction supplies extra obligatory rungs on the ladder.
5. The bounds of the ladder (a_0, R, a_{max}, b_{min}, b_{max}) are exactly computable.
6. 1682 is the boundary case just before the threshold Γ(10) = 1683.
7. The transition 9000 to 9001 is a fractal anchor.
8. Leap and 9-cycle combine to 36 years, fully 3600 years.
9. The Matryoshka structure gives the representations a deep, formal additivity.
10. The classification is empirically confirmed for moduli 2 through 900 and m = 1 through 10 (599 : 200 : 100).
11. The theory is general. It works for every modulus q with p ≡ 1 (mod q).
12. Symmetry knows two forms: full symmetry (a = b) and digital symmetry (dr(a) = dr(b)).

Chapter 13: limitations and further research

This investigation is restricted to the decimal system. In other bases the modulus changes.

The leap analysis is restricted to the Gregorian calendar.

For a_{max} only the ladder route is valid. Direct division via ⌊N/p⌋ is not a valid method when the residue class causes that value not to be a valid representation.

The class assignment of a quarter depends on the chosen definition: 90 days is Class C, 91 days is Class A.

The connections in Chapter 10 are observations. Whether they necessarily follow from the structure or arise from the chosen time units remains an open question.

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